

ACTIVITY 2 — FACILITATOR GUIDE

Lean, Six Sigma, Waste, Voice of Customer, and Value

TYPE	DURATION	DMAIC PHASE	MAPS TO
Core Lab	45 min	DEFINE	K1, A2

Purpose of this activity

Apply Lean and Six Sigma concepts — value, waste, defects, variation (K1, A2).

Separate what the customer actually values from the waste that surrounds it. Capture the Voice of the Customer, translate it into measurable CTQ requirements, and run a waste walk using the eight wastes (DOWNTIME).

What learners produce

A VOC-to-CTQ translation table with measurable targets, a value-added analysis and a waste walk log with every observation tagged to one of the eight wastes.

Materials required

- Learner Worksheet (one per learner)
- Checklist (one per group)
- A laptop with Excel, Google Sheets or LibreOffice Calc
- The CSV files in this activity's data/ and templates/ folders
- Flipchart or A3 sheet and markers
- The course slides for the scenario and the framework

Set-up

Groups of 3-5. Distribute one worksheet per learner and one checklist per group. Every activity runs on the same continuous Contoso Service Desk scenario, so remind learners where this one sits in that story before starting.

Scenario / briefing material

Contoso Service Desk handles employee IT requests. Employees complain that tickets take too long to be assigned, that they must keep chasing for status, and that agents apply different definitions of what counts as "resolved". Handoffs between the front line and the specialist queues are unclear, and although ticket data exists it is inconsistent. You are supporting the improvement team as a Yellow Belt — you do not lead the project.

Two weeks of data have been collected: 400 tickets, of which 96 contained at least one defect, with 120 defects recorded in total across 6 defect opportunities per ticket. Mean assignment time is 55 minutes (median 46) against a 30-minute improvement goal. The CSV files in this activity's data/ folder are that extract.

Files in this activity folder

data/ — the mock data you analyse:

- voice-of-customer.csv — 10 rows (id, customer_statement, source, date)
- waste-walk-observations.csv — 10 rows (obs_id, observation, process_step, time_lost_min, waste_type)

templates/ — the worksheets you complete:

- value-added-analysis.csv (step_no, activity, VA_BVA_NVA, justification)
- voc-to-ctq.csv (voc_id, customer_statement, need, ctq_requirement, measure, target)

Open these in Excel, Google Sheets or LibreOffice Calc. Work on a copy so the originals stay clean for revision.

Tag every waste-walk observation with one of the eight DOWNTIME waste types.

How to run it — step by step

#	Instruction to learners	Facilitator note
1	Capture the Voice of the Customer — record at least five verbatim customer statements.	The classic error is a CTQ that is not measurable ('better communication') — insist on a number.
2	Translate each VOC statement into a need, then into a measurable CTQ requirement.	Learners confuse defect and waste; hold the line that a defect fails a CTQ.
3	Classify each process activity as value-added, business-value-added or non-value-added.	Watch for everything being labelled NVA — approvals are often BVA, not pure waste.
4	Conduct a waste walk and tag each observation against the eight wastes (DOWNTIME).	
5	Distinguish a defect (output fails CTQ) from waste (effort the customer will not pay for).	
6	Identify which single waste type appears most often in your scenario.	

Suggested timing

Minutes	Phase
5 min	Read the VOC data
12 min	Translate VOC to needs and CTQs
10 min	Classify activities as VA / BVA / NVA
13 min	Run the waste walk and tag DOWNTIME
5 min	Identify the dominant waste type

Total: 45 minutes

Discussion & decision prompts

1. Which VOC statement was hardest to turn into something measurable, and why?
2. Where did you disagree about value-added versus business-value-added?
3. Which waste type appears most often — and what does that suggest about the cause?

Debrief

Bring the class back together and draw out the learning:

- Ask two or three groups to show what they produced, not just what they concluded.
- Ask what surprised them — the gap between what they assumed and what the data showed.
- Link the activity explicitly back to the DEFINE phase of DMAIC.
- Remind learners this activity rehearses what the Case Study assessment will require.

Reflection questions:

- Which of the eight wastes is most visible in your own work this week?
- What does your customer actually value that you have never measured?

Facilitator notes — what to watch for

- The classic error is a CTQ that is not measurable ('better communication') — insist on a number.
- Learners confuse defect and waste; hold the line that a defect fails a CTQ.
- Watch for everything being labelled NVA — approvals are often BVA, not pure waste.

Success criterion

Every CTQ is measurable with a target, and each waste observation is tagged to one of the eight waste types.

Assessment link

This activity builds the DEFINE phase of DMAIC. All activities are in-class activities assessed indirectly through the open-book Case Study (CS), which reuses the same techniques on the same Contoso Service Desk scenario and the same figures (400 tickets, yield 76%, DPMO 50,000). Knowledge is assessed in the Written Assessment (WA-SAQ). Nothing is assessed that is not practised here or taught in the slides.